

PreviewCodeBlame

Raw
CopyDownload
Edit

```
In [ ]: import pandas as pd
import numpy as np
from datetime import datetime

#create a date variable that contains the dates
dates = [datetime(2011, 1, 2),
datetime(2011, 1, 5),
datetime(2011, 1, 7),
datetime(2011, 1, 8),
datetime(2011, 1, 10),
datetime(2011, 1, 12)]

#with this date object, create a series
ts = pd.Series(np.random.randn(6), index=dates)
print("Date Index \n",ts)
```

Date Index
2011-01-02 -0.074808
2011-01-05 -0.499103
2011-01-07 0.695104
2011-01-08 -0.342038
2011-01-10 -0.418703
2011-01-12 -0.364005
dtype: float64

```
In [ ]: print("\nDate Index alone from the series: \n",ts.index)
```

Date Index alone from the series:
DatetimeIndex(['2011-01-02', '2011-01-05', '2011-01-07', '2011-01-08',
'2011-01-10', '2011-01-12'],
dtype='datetime64[ns]', freq=None)

```
In [ ]: stamp=ts.index[2]
print("\nSecond index:\n", stamp)
```

Second index:
2011-01-07 00:00:00

```
In [ ]: #play like a ball with long-times series
longer_ts=pd.Series(np.random.randn(1000),
index=pd.date_range("2000-01-01",periods=1000))
print("\nDaterange up to 1000 days:\n",longer_ts)
```

Daterange up to 1000 days:
2000-01-01 0.034686
2000-01-02 0.665229
2000-01-03 -0.890827
2000-01-04 0.279958
2000-01-05 -0.949267
...
2002-09-22 0.147612
2002-09-23 0.146094
2002-09-24 -0.733111
2002-09-25 -0.391483
2002-09-26 -0.719914
Freq: D, Length: 1000, dtype: float64

```
In [ ]: print("\nDate range-2001 year:\n", longer_ts["2001"])
```

Date range-2001 year:
2001-01-01 -1.575795
2001-01-02 0.687294
2001-01-03 2.044191
2001-01-04 -0.016344
2001-01-05 -0.321923
...
2001-12-27 -0.035727
2001-12-28 -1.398059
2001-12-29 -0.716053
2001-12-30 -1.311292
2001-12-31 -1.743739
Freq: D, Length: 365, dtype: float64

```
In [ ]: print("\nDate range-05 month 2001 year:\n", longer_ts["2001-05"])
```

Date range-05 month 2001 year:
2001-05-01 -0.642660
2001-05-02 -0.770738
2001-05-03 -1.606449
2001-05-04 0.642911
2001-05-05 -0.963976
2001-05-06 -0.954181
2001-05-07 -0.796744
2001-05-08 0.710393
2001-05-09 -0.183788
2001-05-10 1.229141
2001-05-11 -0.499416
2001-05-12 1.264031
2001-05-13 1.805908
2001-05-14 1.363706
2001-05-15 -0.319850

```
2001-05-16    -0.110340
2001-05-17    -1.230445
2001-05-18     0.993718
2001-05-19     0.654259
2001-05-20    -1.666631
2001-05-21     1.129996
2001-05-22    -0.051029
2001-05-23     0.175527
2001-05-24     0.420804
2001-05-25    -0.228889
2001-05-26    -1.756085
2001-05-27     0.313385
2001-05-28    -1.198183
2001-05-29    -0.000052
2001-05-30     0.501831
2001-05-31     1.121728
Freq: D, dtype: float64
```

```
In [ ]: print("\nDate range-first few dates in 2000 year:\n",longer_ts["2000"].head())
```

```
Date range-first few dates in 2000 year:
2000-01-01     0.034686
2000-01-02     0.665229
2000-01-03    -0.890827
2000-01-04     0.279958
2000-01-05    -0.949267
Freq: D, dtype: float64
```

```
In [ ]: print("\nDate range-first 15 dates in 02 month 2001 year:\n",longer_ts["2000-02"].head(15))
```

```
Date range-first 15 dates in 02 month 2001 year:
2000-02-01    -0.044375
2000-02-02     1.374531
2000-02-03    -0.809166
2000-02-04    -0.972864
2000-02-05     1.703179
2000-02-06    -0.304608
2000-02-07     0.429192
2000-02-08    -1.217413
2000-02-09    -0.242821
2000-02-10    -1.104093
2000-02-11     0.792972
2000-02-12     1.050004
2000-02-13    -0.803575
2000-02-14    -0.049461
2000-02-15     1.859175
Freq: D, dtype: float64
```

```
In [ ]: print("\nDate range-rom 26-9-2001:\n",longer_ts[datetime(2000,9,26):])
```

```
Date range-rom 26-9-2001:
2000-09-26    -0.219467
2000-09-27     0.001434
2000-09-28     0.073724
2000-09-29     1.806545
2000-09-30    -0.246865
...
2002-09-22     0.147612
2002-09-23     0.146094
2002-09-24    -0.733111
2002-09-25    -0.391483
2002-09-26    -0.719914
Freq: D, Length: 731, dtype: float64
```

```
In [ ]: print("\nDate range from 26-9-2001 to 25-9-2002:\n",longer_ts[datetime(2001,9,26):datetime(2002,9,25)])
```

```
Date range from 26-9-2001 to 25-9-2002:
2001-09-26     0.339419
2001-09-27    -1.320885
2001-09-28     0.513673
2001-09-29     0.724614
2001-09-30     0.825734
...
2002-09-21     0.414729
2002-09-22     0.147612
2002-09-23     0.146094
2002-09-24    -0.733111
2002-09-25    -0.391483
Freq: D, Length: 365, dtype: float64
```